

Product brief — Kill the Quote Spreadsheet (RFx Crew)

Prototype: <https://kill-the-quote-spreadsheet-lac.vercel.app/> **Repo:** <https://github.com/Anirudhmen94/Aerchainfinal> · app_crew: app **Category demo:** Corrugated packaging · Chakan (Pune) snacks plant · 5 vendors · 30 line items · 8 questionnaire items · FX 83.50 INR/USD

1. Problem statement

Category buyers still spend days turning vendor chaos into a spreadsheet. They draft an RFx, email five suppliers, and get back whatever the supplier prefers: an Excel that ignores the template, a PDF with the discount in a footnote, a Word paragraph of commercials, a phone photo of a rate card, or a one-line email (“₹42/kg for the 5-ply... rest same as last year, freight extra”). Retyping takes three days. One VP question — *cheapest per line, only among vendors who cleared quality* — takes a fourth.

We haven’t accepted that as the cost of doing business. This prototype drafts the RFx with an AI co-pilot, accepts replies in any shape, normalizes them into one INR side-by-side matrix with questionnaire answers alongside the numbers, and lets the buyer interrogate the result in plain language through to a defensible award — with uncertainty and evidence treated as first-class product, not footnotes.

2. User groups

Persona	What they need from this system
Category buyer (primary)	Draft fast, see every vendor on one screen, trust every cell, ask “what if we split?”, freeze an award they can defend.
Category / sourcing manager	Visibility when a buyer wants a Fail/incomplete vendor on a line — stub manager notify in Outbox (no Approve panel on Award); Audit trail of freezes and overrides.
Vendor (indirect)	Reply however they like; nobody is forced into a portal template. Award notices (stub) go out only to confirmed winners.
Reviewer / demo driver	Public durable URL, free tabs, seeded ugly-edge dataset, no local setup required for the happy path.

3. Features

Free tabbed workspace — **Draft | Outbox | Inbox | Compare | Ask | Award | Audit** (no unlock gates; Home has no **E2E** button).

Tab	Capability
Draft	Plain-language brief → generate ~30 lines + 8-question quality questionnaire (ISO / FSC / food-contact knockouts) + 5 Indian vendors; edit title/scope/terms/lines.
Outbox	Cover-email previews → stub Send to vendors → data/outbox/. Later: award notices + manager notify stubs (filter: RFQ / Award notice / Manager).
Inbox	Seed/upload the five ugly formats → Parse / Parse all → quotations + questionnaire

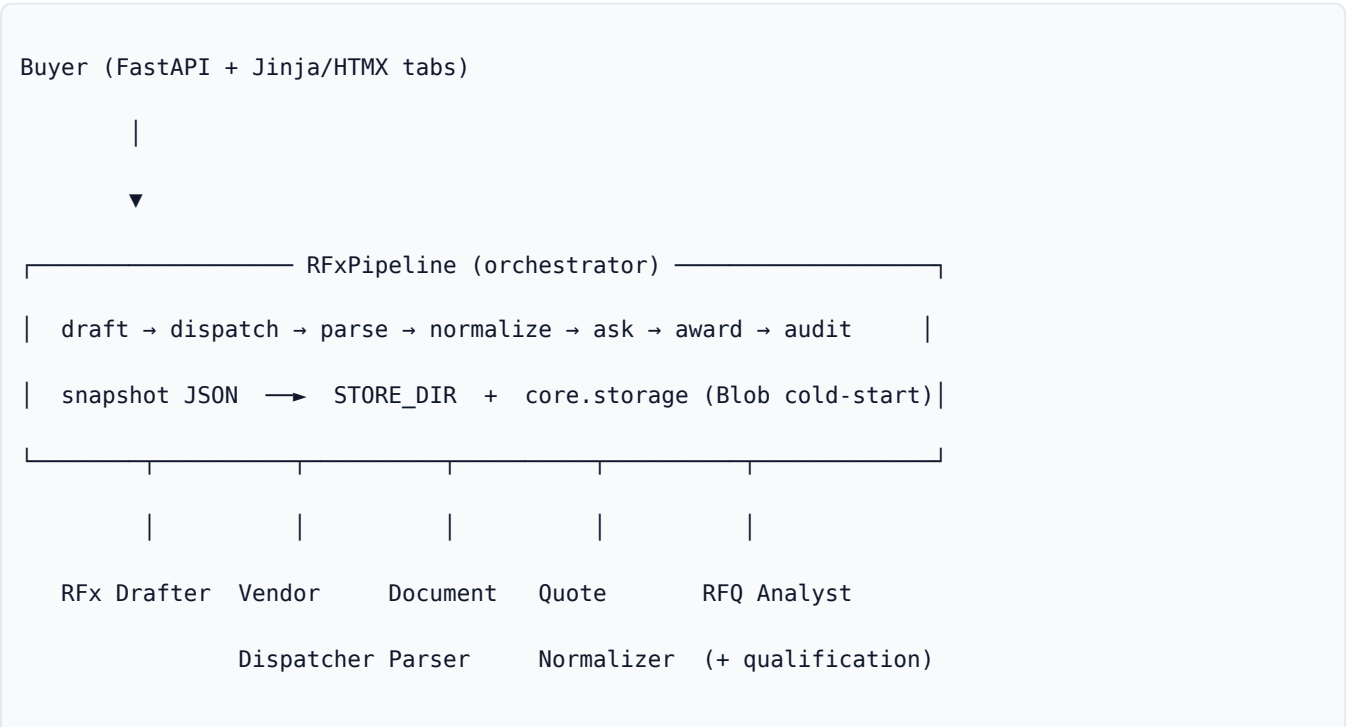
	answers; cards stay quote-aligned with Compare.				
Compare	Side-by-side INR matrix; cell statuses `ok\`	converted\`	uncertain\`	uom_mismatch\`	missing`; knockout Q / XQ badges; source-of-truth evidence drawer (snippet + original file/media).
Ask	Persistent live chat; human tone (senior buyer briefing a colleague); premade chips incl. VP-defend; free-ask tool loop over the frozen matrix.				
Award	Per-line dropdowns (Pass + priced only); suggest cheapest split; partial/override requests notify manager via Outbox (no Approve UI here); freeze; stub award notices; export Excel / CSV / Markdown.				
Audit	Chronological review_log: awards saved, freezes, notices, cell overrides, partial requests.				

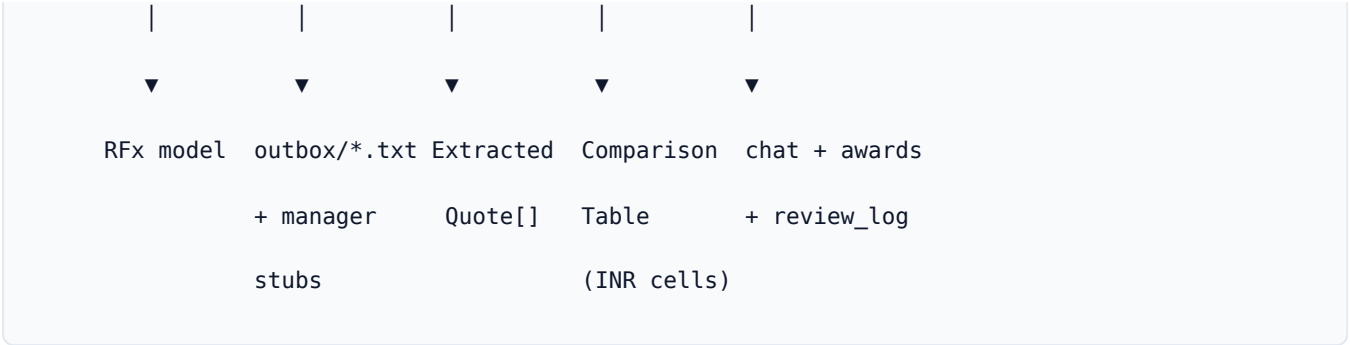
Ugly-edge seed set (data/vendor_responses/): V01 Excel per-100 · V02 PDF ~27/30 + footnote discount · V03 Word USD/1000 · V04 angled PNG per-box/bundle · V05 one-line email ₹/kg + “same as last year”.

Trust mechanics: evidence-linked cells · deterministic FX/UOM · knockout gating · Blob-backed cold-start reload of the full session snapshot.

4. Tech architecture — five bots + orchestrator sync

Agents are plain Python modules with **shared Pydantic contracts** in shared_models.py. There is **no CrewAI/LangChain dependency**. A sequential orchestrator (orchestrator/pipeline.py → RFxPipeline) owns the session, calls each agent’s public API, and **persists a JSON snapshot** after every step (local data/store/ or /tmp on Vercel, dual-written to **Vercel Blob** when BLOB_READ_WRITE_TOKEN is set).





How they hand off / sync

Handoff	Mechanism
Drafter → rest	Emits RFX (lines, questionnaire, vendors) into pipeline state.
Dispatcher → Outbox	Writes stub cover emails under data/outbox/; <code>dispatch_log</code> on the snapshot.
Parser → Normalizer	<code>ExtractedQuote[]</code> (lines, questionnaire answers, confidence, <code>raw_evidence</code>). Companion <code>*.extract.json</code> keeps offline demos honest.
Normalizer → Compare/Ask/Award	Pure transform → <code>ComparisonTable</code> cells + <code>VendorQualification / award_eligible_vendors</code> . No prose.
Analyst → Ask/Award	Reads the frozen comparison; ranking/totals in code; language over tables; <code>validate_award / suggest_split_award</code> .
Award → Outbox / Audit	Manager notify + award notices as outbox stubs; events appended to <code>review_log</code> .
Cross-instance sync	<code>_persist()</code> / <code>load_pipeline()</code> — local JSON first, then Blob — so cold starts recover the same <code>rfx_id</code> .

UI ↔ **agents**: `app_crew.py` routes map 1:1 to pipeline methods (`draft`, `dispatch`, `seed_inbox`/`parse_*`, `normalize`, `ask`, `save_awards`, `freeze/notify`). Tab key send is labelled **Outbox** in the UI.

5. Success metrics

Demo / take-home bar (what “working” means for a reviewer):

Metric	Target / signal
Ugly-edge coverage	All 5 seeded formats parse into the matrix without forcing a template.
Normalization honesty	USD and non-piece UOMs show as converted (or <code>uom_mismatch / uncertain</code> when unsafe); partial coverage shows as missing (e.g. ~27/30).
Trust surface	Every priced cell has openable source-of-truth evidence; Audit shows freeze / override / notice events.
Quality-gated award	Fail/XQ vendors excluded from default shortlist; partial path creates manager Outbox stub (not silent eligibility).
Analyst defensibility	VP-defend / cheapest-qualified-split answers cite computed tables + caveats; free-ask is not hardcoded.
Demo latency & durability	Stub dispatch / structured parse near-instant; Blob cold-start reload of the same <code>rfx_id</code> succeeds (<code>/healthz → "app": "rfx-crew"</code>).
Buyer time (product thesis)	Collapse “retype week + VP question afternoon” into a single live workspace session for a 5×30 event.

Deliberately out of scope for v1 metrics: real email delivery SLAs, vendor portal adoption, ERP post-award sync, multi-tenant ACL.